

Pixel-Level Contrastive Pretrainer for Industrial Image Representation

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Abstract—Industrial quality inspection aims to identify defective parts in industrial production processes. Commonly used methods for industrial quality inspection rely on feature representations that have been pretrained on natural image datasets, such as ImageNet. However, these pretrained models are not specifically tailored for industrial scenarios and therefore do not transfer well to downstream industrial tasks. In this study, we have curated a large-scale industrial production dataset called Ind-2M, which is specifically collected from industrial scenarios. This dataset serves to enhance the industrial representation of pretraining models. Additionally, we propose a Pixel-level COntRastive (PiCO) pretrainer for industrial image representation. PiCO not only improves the global industrial representation through industrial production classification, but also enhances the local industrial representation through pixel-level self-supervision. Experimental results demonstrate that PiCO effectively transfers to downstream industrial tasks, such as multilabel defect classification and anomaly detection, outperforming existing pretrained methods. We hope PiCO can initiate a new paradigm for industrial image pretraining.

Index Terms—Ind-2M, industrial image representation, Pixel-level COntRastive (PiCO), pixel-level contrastive learning, pretrainer.

I. INTRODUCTION

INDUSTRIAL quality inspection is a critical endeavor that seeks to identify and evaluate defects within the realm of industrial manufacturing [1]. Nonetheless, this process is confronted with inherent limitations, such as a scarcity of proficient inspectors and the substantial time investment required to thoroughly examine industrial productions. Consequently, there is a pressing need for automated systems capable of detecting defects within industrial contexts. Furthermore,

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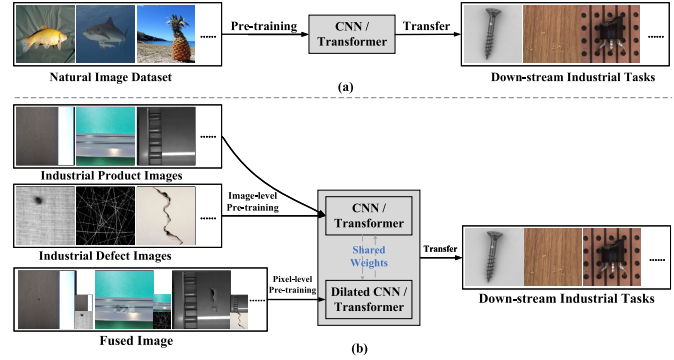


Fig. 1. Pretraining paradigm for industrial image representations. (a) Existing natural image pretraining paradigm learns the common representations from the natural image dataset and transfers to the downstream industrial tasks directly. (b) PiCO pretraining paradigm learns the image-level industrial representations from the large-scale industrial dataset Ind-2M and the pixel-level representations from the fused images, and then transfers to the downstream industrial tasks.

automated industrial quality inspection represents a crucial research area within the instruments and measurements field. It plays a significant role in facilitating the advancement of surface defect detection technologies, such as Strip Steel Surface Defect [2], [3], [4], [5], glass defect detection [6], [7], pavement defect detection [8], [9], [10], LED defect detection, as well as improve industrial anomaly detection technologies [11], [12], [13]. This leads to the overall improvement of various industrial quality inspection practices.

In recent years, the field of industrial defect detection has garnered significant attention, as evidenced by the growing body of research exploring various approaches. These include supervised defect detection and segmentation techniques [14], [15], unsupervised anomaly detection methods [16], [17], and self-supervised anomaly generation approaches [18], [19]. Although considerable advancements have been made in defect detection methods over the past few decades, it is noteworthy that existing techniques commonly rely on feature representations pretrained on ImageNet [20], a natural image dataset. These pretrained features are then directly transferred to downstream industrial tasks, as depicted in Fig. 1(a).

The pretraining of features using natural image datasets may not adequately capture the representation needed for industrial applications for several reasons. First, natural image datasets lack images from real industrial production settings, which hinder their ability to effectively represent industrial scenarios. Second, critical features required for tasks such as defect detection or anomaly detection in industrial settings



Fig. 2. Examples of **Ind-2M** dataset. Left: image examples of industrial products. Right: image examples of industrial defects.

are often localized and small, whereas features in natural image datasets tend to be global and large in scale. This disparity poses a challenge in utilizing natural image datasets for industrial representation. Third, in industrial products, the regions exhibiting defects are often similar in appearance to the regions without defects, making it difficult to distinguish between defective and non-defective features. On the other hand, natural image datasets typically consist of objects that are more easily distinguishable.

To address the aforementioned challenges associated with pretraining models for industrial applications, we have curated a comprehensive dataset called Ind-2M. This dataset has been specifically collected from the industrial scenarios, as depicted in Fig. 2. Ind-2M comprises a vast collection of 2.2 million industrial images, sourced from both publicly available industrial datasets and data obtained through web crawling. The dataset encompasses 1.6 million images of non-defective industrial products, as well as 0.6 million images depicting various defects found in industrial settings. By leveraging the Ind-2M dataset, our objective is to facilitate the advancement of industrial representation through pretraining models.

We additionally propose a Pixel-level COntrastive (PiCO) pretrainer for enhancing industrial image representation, as illustrated in Fig. 1(b). PiCO is designed to acquire comprehensive industrial representations by performing classification tasks on industrial products. The labels for these tasks are obtained either from data gathered from the internet or from the product types within the dataset. Furthermore, given that the Ind-2M dataset includes 0.6 million defect images obtained from internet sources, PiCO is capable of learning defect-related features through the classification of industrial defect images.

Moreover, acquiring a sufficient number of defective product images can be challenging. To address this issue, we present a heuristic data generation approach that allows us to generate a substantial quantity of defective product images. This method leverages the classification activation mapping (CAM) [21] for product and defect classification within PiCO. By utilizing CAM, we can extract the approximate regions of both the products and defects. Subsequently, we employ Poisson Blending [22] to seamlessly merge these product

regions with the corresponding defective regions, resulting in realistic defective product images.

In order to obtain the defective regions using the CAM technique, our proposed method, PiCO, incorporates a hybrid triplet mining loss to constrain the pixel-level representation. This constraint is achieved through supervised feature learning. Specifically, when the same defect is inserted into different industrial product images, the feature distance between defects within these images should be smaller than the feature distance between defects and industrial products. During the pretraining of the convolutional neural network (CNN) model, we modify the architecture by abandoning the last stride of the CNN and instead using the dilated CNN approach, as proposed by [23]. This modification allows us to recover the receptive field, thereby enabling the industrial feature representations to retain more detailed features. PiCO can be also applied to vision transformer (ViT) [24] without modification. Furthermore, we constrain the pixel-level representation of the features obtained from either the dilated CNN or ViT.

In our experimental evaluation, we assess the performance of PiCO on downstream industrial tasks, specifically multilabel defect classification and anomaly detection. Our results demonstrate that PiCO surpasses existing pretrained methods in terms of performance. Notably, pretraining method PiCO with the large-scale industrial dataset Ind-2M yields a significant improvement compared to previous pretrained models based on ImageNet.

The primary motivation driving this research stems from the inadequacy of existing pretraining methods for industrial intelligent quality inspection. Consequently, we address this issue by introducing the Ind-2M dataset, a comprehensive large-scale dataset specifically designed for pretraining large-scale industrial models. Furthermore, we propose PiCO, a novel approach tailored for pretraining these models. In summary, the main contributions of this work can be summarized as follows.

1. We have curated a large-scale industrial dataset, Ind-2M, specifically designed to capture real-world industrial scenarios. To the best of our knowledge, Ind-2M represents the first comprehensive industrial dataset for

pretraining industrial image models. Ind-2M dataset is available at <https://github.com/BingkeZhu/Ind-2M>.

- 2) In this work, we introduce PiCO, a pixel-level contrastive pretrainer that aims to enhance industrial image representations. PiCO incorporates both global and local representations, employs a heuristic data generation approach, and leverages pixel-level features. Notably, PiCO is model-agnostic, capable of benefiting both CNN and ViT models, thereby facilitating improved industrial representations and enhancing performance on downstream industrial tasks.
- 3) Through extensive experimentation, we demonstrate that both the Ind-2M dataset and the PiCO pretrainer yield significant improvements in industrial representations compared to previous models pretrained on ImageNet. We believe that the introduction of the Ind-2M dataset and the PiCO pretrainer marks the beginning of a new era in industrial image pretraining, fostering advancements in this domain.

II. RELATED WORK

A. Industrial Defect Detection

Industrial quality inspection is an important step in industrial production. Automatic industrial defect detection can reduce time costs and labor costs effectively, and thus receives much research attention. Feature collection and compression network [25] adopt the compressed multiscale feature merging for real-time defect detection. The attention-guided segmentation network [14] combines UNet and biConvLSTM to improve the pixel-level welding defect detection performance. An encoder-decoder structure [26] is proposed to fuse multibranch and multiscale features for surface defect detection. Also, a fake-to-real domain transformation GAN is used to transform the rendered samples from the fake domain into the real defect domain. DCAM-Net [27] proposes a rapid detection network based on deformable convolution and attention mechanism. It introduces CLAHE for data augmentation, an EDE-block for capturing complex and irregular defects, and a CA module for improved defect localization. MANet [8] utilizes the encoder-decoder architecture with MobileNet as the backbone network for feature extraction. Multiscale convolution kernels are employed in depthwise separable convolution (DSCov) layers, and a hybrid attention mechanism is incorporated to infer spatial points and interchannel relationship features. NDP-Net [28] utilizes an encoding module to extract multiscale discriminative features and incorporates artificial defects and a pixel-level defect perception loss to enhance defect detection. Additionally, a reference-based attention module and a multiscale defect segmentation module are introduced for precise defect detection and segmentation. MRD-Net [29] is a multiscale feature enhancement fusion and reverse attention network for real-time and end-to-end defect segmentation.

In the past, these industrial defect detection methods were based on fine-tuning a network pretrained on ImageNet using specific industrial defect detection task data. However, this

article questions the suitability of this ImageNet pretraining approach for industrial defect detection tasks. To address this issue, this article proposes a large-scale pretraining dataset specifically designed for industrial defect detection tasks. Additionally, a pixel-level contrastive learning-based pretraining method is introduced for this dataset, aiming to enhance the model's pretraining feature representation capability for industrial defect detection tasks and consequently improve accuracy in downstream industrial defect detection tasks.

B. Anomaly Detection

Recently, anomaly detection has raised a large amount of interest for unsupervised defect detection [16], [30], [31]. Anomaly detection trains only on the normal images, and detects the defects on the anomaly images, thus reduces the labeling costs. PatchCore [16] uses a maximally representative memory bank of nominal patch features. PaDim [17] makes use of a pretrained CNN for patch embedding, and of multivariate Gaussian distributions to get a probabilistic representation of the normal class. SSPCAB [32] proposes to integrate the reconstruction-based functionality into a self-supervised predictive architectural building block. Reverse distillation [33] proposes a T-S model consisting of a teacher encoder and a student decoder and introduce a simple yet effective "reverse distillation" paradigm accordingly. Recently, uniAD [30] is proposed to accomplish anomaly detection for multiple classes with a unified framework with a novel transformer-based decoder. OTB-AAE [34] proposes an anomaly detection framework based on adversarial autoencoder (AAE) to capture the normality distribution of high-dimensional images in industry and introduces the output-turn-back structure (OTB) and combine $L1$ loss and structural similarity (SSIM) loss functions to enhance the discriminant capability of the discriminator. SMCC [35] utilizes a pretrained model for feature extraction and employs a Gaussian mixture model to cluster and obtain cluster centers. This enables better discrimination between normal and abnormal sample features, as normal sample features are tightly distributed around the cluster centers. By leveraging competition among different tasks in a self-supervised multitask setting, MTHM [36] learns compact and efficient representations for detection tasks. Additionally, introducing other semantic tasks allows the shared encoder to learn beyond image reconstruction, improving the model's reconstruction capability for anomalies.

Similar to industrial defect detection tasks, these industrial anomaly detection methods also rely on models pretrained on ImageNet. They extract features from the ImageNet pretrained models for feature transformation, enabling the identification of abnormal regions. It is worth noting that ImageNet pretrained features are particularly important in anomaly detection tasks. Fine-tuning the pretrained ImageNet portion of the model can lead to a drastic decrease in performance [30]. Therefore, designing a pretraining method specifically suitable for industrial anomaly detection tasks is of great research significance in order to improve the performance of industrial anomaly detection models.

C. Network Pretraining

ImageNet [20] pretraining is the most common network pretraining method for image representation [37], [38], [39], [40], [41]. Recently, the contrastive learning-based self-supervised methods like MoCos [42], [43] have attracted much attention. Also, the mask reconstruction-based self-supervised methods like MAE [44] improve the image representations of ViT largely. SimCLR [45] simplifies contrastive self-supervised learning algorithms without requiring specialized architectures or a memory bank. SimSiam [46] provides a hypothesis on the implication of stop-gradient, and further shows proof-of-concept experiments. SimMIM [47] simplifies MAE-based methods without special designs such as blockwise masking and tokenization via discrete VAE or clustering.

The existing network pretraining methods are designed for general object recognition tasks. However, these methods do not take into account the characteristics of industrial defect detection tasks. For example, these methods are pretrained on the ImageNet dataset, resulting in features that lack understanding of industrial products and defects, such as the inability to understand scratches. Additionally, these methods are focused on large target areas with prominent features in the images. However, they may not yield significant results for industrial defects that are less obvious in the image, such as cracks. Therefore, there is a need to research a pretraining method specifically tailored for industrial feature representation tasks.

Existing methods have failed to address the pretraining paradigms that are suitable for intelligent quality inspection in industrial settings for the following reasons: first, obtaining industrial data is challenging because the production environment typically contains a substantial number of normal samples to ensure production quality, but lacks defect samples. Second, the defect area in industrial images is often small compared to the entire image, whereas commonly used general intelligence recognition datasets like ImageNet primarily consist of images where the target occupies a significant area. Lastly, current methods primarily focus on training the entire image and do not incorporate pixel-level differential contrastive learning, which poses difficulties in accurately identifying defects. In contrast to previous approaches, our work specifically targets industrial scenarios. To address the issue of difficult access to industrial data, we introduce a large-scale Ind-2M dataset and propose a synthetic defect generation method to augment a large number of defect training samples. To tackle the problems of small defect areas and the lack of pixel-level differential constraints, this article introduces pixel-level contrastive learning as a solution.

III. METHOD

Fig. 3 provides an overview of the PiCO framework, which encompasses three distinct stages in each iteration, namely the image classification stage, the image blending stage, and the pixel-level representations stage.

In image classification stage, PiCO classifies the product images and defect images with predefined product and defect

categories. Then, we can obtain the CAM for product regions and defect regions. In image blending stage, PiCO fuses the industrial product images and defect images using Poisson Blending, and the previous CAM can be used to guide the blending process. In pixel-level representations stage, the previous classifying CNN removes the last striding, and uses dilation convolution [23] to remedy the receptive field. Then, PiCO optimizes the pixel-level representations by the proposed hybrid triplet mining.

In this section, we will first elaborate on the large-scale industrial image dataset, Ind-2M dataset. Then, we will introduce three stages of PiCO, respectively. Finally, we will present the overall pipeline of PiCO.

A. Ind-2M Dataset

In order to acquire robust image representations for industrial applications, we have curated a comprehensive dataset known as Ind-2M. This dataset is specifically tailored for industrial scenarios and comprises a vast collection of 2.2 million industrial images. The images within Ind-2M are sourced from both publicly available industrial datasets and data obtained through web crawling. Notably, the dataset consists of 1.6 million product images and 0.6 million defect images.

As shown in Table I, the Ind-2M dataset consists of 18 publicly available datasets, 31 web-crawled industrial product scene data, and 30 web-crawled industrial defect scene data. The open-source datasets include SewerML [15], Aluminum Profile [48], Kylberg [49], NEU surface defect database [2], Severstal [50], DAGM 2007 [51], RSDDs [52], KTH-TIPS [53], PKU-Market-PCB [54], AITEX [55], elpv-dataset [56], KolektorSDD [57], Deep-PCB [58], SDNET2018 [59], Fabric Defect [60], Insulator Defect Image Dataset [61], BSDData [62], and TILDA Textile Texture-Database [63] datasets. These datasets cover 11 different scenes including pipeline damage, aluminum profiles, textile industry, steel strip, magnetic tiles, railway tracks, concrete, printed circuit boards, solar panels, metal, and wires. Each dataset typically focuses on a specific product category. To enable PiCO classification, we annotate the images in each dataset with their corresponding product category labels.

Additionally, we employ web crawling techniques to acquire industrial product images from the internet. We meticulously amassed a comprehensive dataset of 221 062 images sourced from a variety of industrial e-commerce websites, industrial image websites, and similar platforms. These images were acquired by employing 31 carefully selected keywords, including *height gauge*, *caliper*, *gloves*, *safety helmet*, *light bulb*, *lighting fixture*, *cable*, *clock*, *shoe covers*, *screen*, *sensor*, *chemicals*, *personal protective equipment*, *transformer*, *inverter*, *security products*, *locator*, *mirror*, *pipe inspection device*, *syringe*, *laboratory equipment*, *power supply*, *instruments*, *tools*, *electrical distribution equipment*, *cleaning tools*, *electrical and electronic tools*, *projector*, and *socket*. The aggregation of these images forms our dataset, which focuses specifically on industrial products. Consequently, the obtained

TABLE I

COMPOSITION OF THE IND-2M DATASET. THE IND-2M DATASET CONSISTS OF 18 PUBLICLY AVAILABLE INDUSTRIAL DATASETS, 31 WEB-CRAWLED INDUSTRIAL PRODUCT SCENE IMAGES, AND 30 WEB-CRAWLED INDUSTRIAL DEFECT SCENE IMAGES

Dataset	#Images	Description
<i>Public Dataset</i>		
SewerML [15]	1,300,000	Sewer-ML is a sewer defect dataset. It contains 1.3 million images, from 75,618 videos collected from three Danish water utility companies over nine years.
Aluminum Profile [48]	10,000	Aluminum Profile Dataset consists of monitoring images of aluminum profiles with defects from actual production, where each image contains one or more types of defects.
Kylberg [49]	4,480	Kylberg dataset contains 28 texture classes, and 160 unique texture patches per class.
NEU surface defect database [2]	1,800	The dataset collected 6 types of defects, including inclusion, scratches, pressed oxide, cracks, pinholes, and patches. Each defect type consists of 300 images, with image dimensions of 200×200 .
Severstal [50]	18,074	The dataset provides four types of surface defects on steel strips. The training set consists of 12,568 images, while the test set has 5,506 images.
DAGM 2007 [51]	11,500	DAGM 2007 dataset is a synthetic dataset for defect detection on textured surfaces.
RSDDs [52]	195	The RSDDs dataset contains two types of datasets: the first is a type I RSDDs dataset captured from the fast lane, which contains 67 challenging images. The second is a Type II RSDDs dataset captured from a normal/heavy transportation track, which contains 128 challenging images.
KTH-TIPS [53]	810	The KTH-TIPS (Textures under varying Illumination, Pose and Scale) image database was created to extend the CURET database in two directions, by providing variations in scale as well as pose and illumination, and by imaging other samples of a subset of its materials in different settings.
PKU-Market-PCB [54]	1,386	Printed Circuit Board (PCB) Defect Dataset. It is a publicly available synthetic PCB dataset consisting of 1,386 images with 6 types of defects (leakage, mouse bite, open, short, stray, and scattered copper) for image detection, classification, and registration tasks.
AITEX [55]	245	The textile fabric database consists of 245 images of 7 different fabrics. There are 140 defect-free images, 20 for each type of fabric.
elpv-dataset [56]	2,624	The dataset contains 2,624 samples of 300×300 pixels 8-bit grayscale images of functional and defective solar cells with varying degree of degradations extracted from 44 different solar modules. The defects in the annotated images are either of intrinsic or extrinsic type and are known to reduce the power efficiency of solar modules.
KolektorSDD [57]	399	The dataset is constructed from images of defective production items. The images were captured in a controlled industrial environment in a real-world case.
DeepPCB [58]	3,000	DeepPCB dataset contains 1,500 image pairs, each of which consists of a defect-free template image and an aligned tested image with annotations including positions of 6 most common types of PCB defects: open, short, mousebite, spur, pin hole, and spurious copper.
SDNET2018 [59]	56,000	The dataset is primarily used to study the detection and extraction algorithms of bridge cracks.
Fabric Defect [60]	20,000	This dataset covers various important defects in the textile industry, including different types of flaws found in fabrics. Each image contains one or more flaws. The dataset includes two categories: plain fabrics and patterned fabrics. The plain fabric data consists of approximately 8,000 images, while the patterned fabric data consists of approximately 12,000 images.
Insulator Defect Image Dataset [61]	848	The Insulator Defect Image Dataset (IDID) consists of labeled high quality images of transmission line insulators. The images have insulator string as the primary subject and parent class.
BSData [62]	1,104	This dataset contains 1,104 images with 3 channels, and includes 394 annotated images for surface damage input pitting.
TILDA Textile Texture-Database [63]	3,254	TILDA is a Textile Texture Database. A total of eight representative textile kinds were included in the database. Based on the analysis of textile atlases, seven error classes were defined.
<i>Web-crawled Dataset</i>		
Product Dataset	221,062	We collected a total of 221,062 images from industrial e-commerce websites, industrial image websites, and other similar platforms. These images were obtained based on 31 keywords such as <i>height gauge, caliper, gloves, safety helmet, light bulb, lighting fixture, cable, clock, shoe covers, screen, sensor, chemicals, personal protective equipment, transformer, inverter, security products, locator, mirror, pipe inspection device, syringe, laboratory equipment, power supply, instruments, tools, electrical distribution equipment, cleaning tools, electrical and electronic tools, projector, and socket</i> . These images comprise our industrial product dataset.
Defect Dataset	614,002	We obtained a total of 614,002 images by crawling the web, based on 30 keywords including <i>cracks, repairs, dents, protrusions, deformations, defects, cuts, holes, fractures, oil stains, wrinkles, loose threads, contamination, extrusions, punctures, worn areas, misalignments, bubbles, blemishes, collapses, scratches, burn marks, adhesions, cracks, looseness, pitting, decay, peelings, mold, and impurities</i> . These images form our industrial defect dataset.

industrial product images are associated with specific product category labels, facilitating subsequent PiCO classification tasks.

Furthermore, we have created the Ind-2M dataset, which encompasses a staggering 0.6 million industrial defect images meticulously obtained through web crawling. Our comprehensive web crawling process allowed us to aggregate a total of 614,002 images, guided by 30 carefully selected keywords. These keywords include *cracks, repairs, dents, protrusions, deformations, defects, cuts, holes, fractures, oil stains, wrinkles, loose threads, contamination, extrusions,*

punctures, worn areas, misalignments, bubbles, blemishes, collapses, scratches, burn marks, adhesions, cracks, looseness, pitting, decay, peelings, mold, and impurities. This extensive collection of images forms our industrially focused defect dataset, encompassing a wide range of defect categories found within the industrial domain.

In order to guarantee the integrity and validity of the collected data through web crawling, we implemented a rigorous validation process. Following the accumulation of a substantial volume of industrial product and defect images, we enlisted the expertise of ten highly skilled data engineers

who possessed professional training in the domain. These individuals were tasked with meticulously reviewing and filtering the crawled data. Their extensive knowledge and understanding of industrial products and defects enabled them to identify and eliminate images that did not align with the designated product names or correspond to the specified defect categories. By employing such manual filtering, we ensured the data's accuracy and adherence to the intended industrial context, contributing to the overall validity of our dataset.

B. Industrial Image Classification

The image classification stage serves two purposes in the PiCO pretrainer. First, it facilitates the acquisition of global semantic understanding capabilities by pretrained models, akin to previous supervised pretraining methods employed in deep learning models. Second, it leverages CAM to obtain approximate positions of industrial products and defects, establishing a positional foundation for the subsequent blending of industrial defect images.

During the industrial image classification stage, the CNN or ViT models are employed to classify the predefined industrial product categories and industrial defect categories using both the industrial product images and defect images. For this purpose, PiCO utilizes two fully connected layers, one for product classification and the other for defect classification. Notably, PiCO ensures that the product images are classified without considering the defect categories, and vice versa. Subsequently, the CAM can be obtained for both the product regions and defect regions by multiplying the category weights with the corresponding feature representations.

Formally, given the input industrial product image I_P and industrial defect image I_D , the classification process can be represented as

$$X_i = F(I_i), i \in \{D, P\} \quad (1)$$

$$C_i = fc_i(X_i), i \in \{D, P\} \quad (2)$$

where X_i denotes the feature representations, F denotes the CNN or transformer networks, fc denotes the fully connected layers, and C denotes the classification categories.

C. Industrial Image Blending

The industrial image blending stage is capable of synthesizing a substantial number of industrial defect images, thereby alleviating the scarcity and challenges associated with acquiring such images. Moreover, it generates masks that facilitate the subsequent learning of pixel-level representations.

In the image blending stage, PiCO employs Poisson blending to fuse industrial product images and defect images. Guided by the previous CAM, the defects are randomly pasted onto the product regions. Initially, the top 50% challenging samples, denoted as S^D , are selected from the defect images in each batch based on the previous defect classification loss $L_{CE}(C_D, gt_D)$. Subsequently, the defect CAM S^D is normalized and binarized using a threshold of 0.5 to generate defect masks. The paste product locations are then randomly chosen based on the normalized CAM probability S^P . For

Algorithm 1 PiCO

Input: Product Image I_A , Product Image I_B , Defect Image I_C , Random Initialized Model F , Iterative Time T

Output: Pre-trained Model F

```

1: for  $t = 1 : T$  do
2:   if  $t \% 2 == 0$  then
3:     Inputting  $I_A, I_B, I_C$  to model  $F$ , update the model
       parameters with cross entropy loss.
4:     Calculating the CAM of products and defects,
       obtaining  $S_A, S_B, S_C$ .
5:   else
6:     Fusing  $I_A$  and  $I_C$  using CAM  $S_A$  and  $S_C$ , with
       Poisson Blending, as well as  $I_B$  and  $I_C$ , obtaining
        $I_A^C$  and  $I_B^C$ .
7:     Using dilated CNN or ViT to obtain the pixel-level
       representations, and using defect masks to split the
       product and defect regions.
8:     Updating model using hybrid triplet mining loss.
9:   end if
10: end for

```

each challenging defect image, the defects are pasted onto the selected locations of two product images using Poisson blending. The fusion consistency of the divergence field is optimized during this process, expressed as follows:

$$\min_f \iint_{\Omega} |\nabla f - \mathbf{v}|^2, \text{ with } f|_{\partial\Omega} = f^*|_{\partial\Omega} \quad (3)$$

where f is the fused image, ∇f is the gradient of f , Ω denotes the defect image, and $\mathbf{v}$ is the gradient of defect image, while f^* denotes the product image.

D. Industrial Pixel-Level Representation

The industrial pixel-level representation stage employs contrastive learning techniques to enhance the discrimination of local detailed features at the pixel level. This enables pretrained models to effectively distinguish industrial products and defects, thereby establishing a strong foundation for downstream industrial intelligent defect detection.

In the pixel-level representation stage, when employing a CNN model for pretraining, the final striding of the CNN is eliminated, and dilation convolution [23] is employed to address the receptive field. Alternatively, when utilizing a ViT as backbone, the vanilla transformer can be used for pixel-level representations, as the total striding of the ViT is 16. It should be noted that the model in the pixel-level representation stage shares weights with the image classification stage.

Both CNN and ViT architectures play crucial roles in industrial applications. Consequently, we propose a pixel-level representation pretraining approach for both CNN and ViT. In CNN architectures, which typically involve five downsampling operations in the spatial dimensions, removing a downsampling operation leads to a reduction in the receptive field for subsequent convolutional operations. To tackle this issue, we employ dilated convolutions in each convolutional layer following the removal of a downsampling

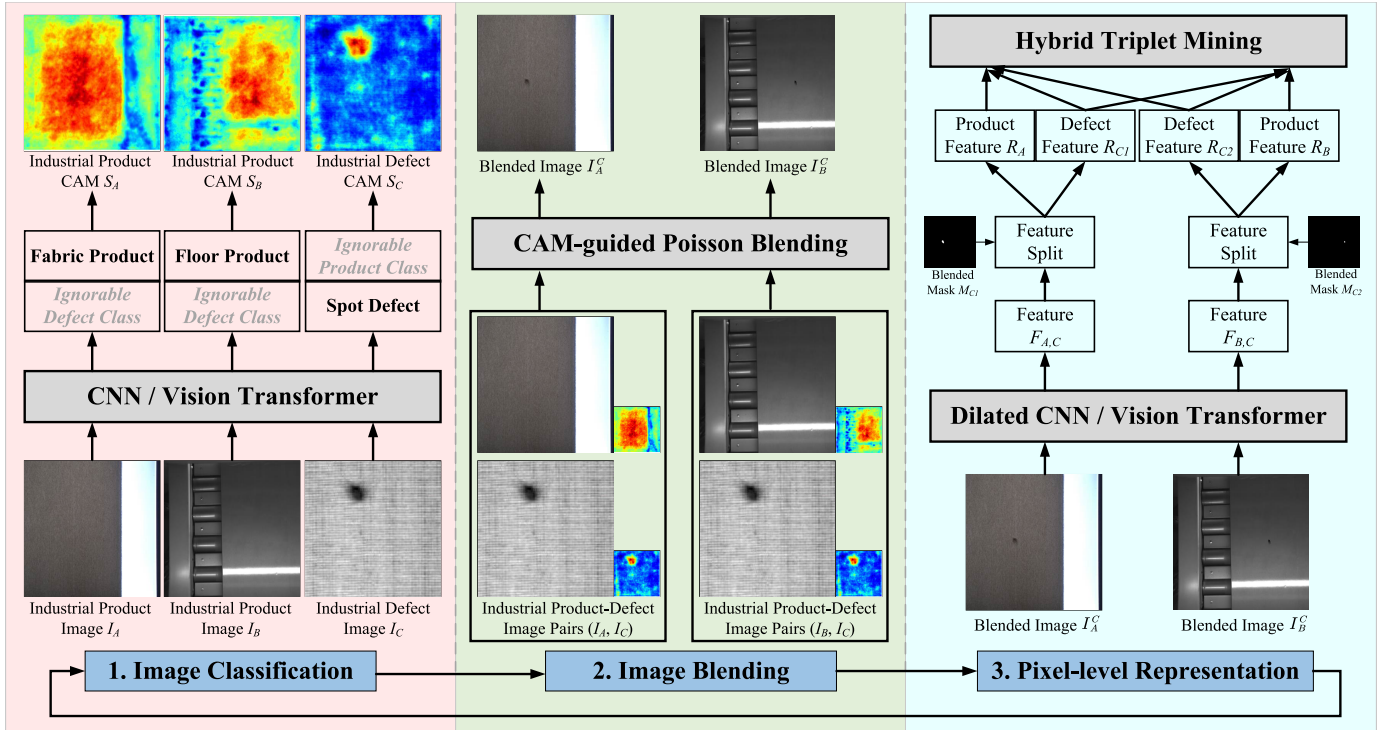


Fig. 3. Framework of PiCO. In **image classification** stage, both of industrial product images and defect images are used to image classification, and obtain the categories of products as well as defects. In **image blending** stage, industrial product images and defect images are fused using CAM-guided Poisson blending. In **pixel-level representations** stage, the blended images are sent into the shared weights dilated CNN or transformer to learn pixel-level representations, optimized by the proposed hybrid triplet mining.

operation to effectively enlarge their receptive fields. For more comprehensive details, please refer to the Deeplab [23]. On the other hand, in the case of ViT, there are no downsampling operations in the spatial dimensions, eliminating the need to remove downsampling operations or employ dilated convolutions to compensate for the receptive field.

Then, PiCO optimizes the pixel-level representations by the proposed hybrid triplet mining. That is, the distance of same defects from different images is smaller than that of defects and products. Formally, given the defective product features $F_{A,C}$ and $F_{B,C}$, we use the blended mask M_{C1} and M_{C2} to split the features, respectively. Then, we constrain the product feature R_A , R_B , and defect feature R_{C1} , R_{C2} with hybrid triplet mining, represented as follows:

$$L_{tri1} = [d(R_{C1}, R_{C2}) - d(R_A, R_{C1}) + \alpha]_+ \quad (4)$$

$$L_{tri2} = [d(R_{C1}, R_{C2}) - d(R_B, R_{C1}) + \alpha]_+ \quad (5)$$

$$L_{tri3} = [d(R_{C1}, R_{C2}) - d(R_A, R_{C2}) + \alpha]_+ \quad (6)$$

$$L_{tri4} = [d(R_{C1}, R_{C2}) - d(R_B, R_{C2}) + \alpha]_+ \quad (7)$$

where $d(\cdot, \cdot)$ is the L2 distance between features, and α is the hyperparameter for triplet loss, which set to 1 by default. Thus, the hybrid triplet loss can be represented as

$$L_{\text{hybrid}} = \lambda_1 \cdot L_{tri1} + \lambda_2 \cdot L_{tri2} + \lambda_3 \cdot L_{tri3} + \lambda_4 \cdot L_{tri4}. \quad (8)$$

In Fig. 4, we illustrate the process of pixel-level representation. We adopt a pixel-wise separation approach in the deep feature space for the same industrial defect observed on two different industrial products. This involves extracting the distinct product features, denoted as R_A and R_B , respectively.

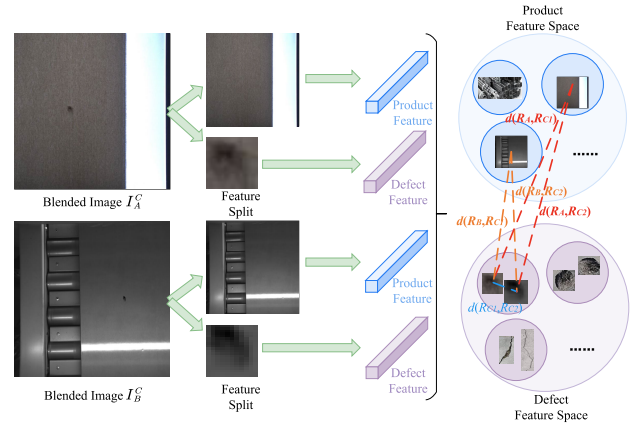


Fig. 4. Illustration of industrial pixel-level representation. For the same industrial defect and two industrial products, we perform pixel-wise separation in the deep feature space, extracting the product features R_A and R_B . The defect features R_{C1} and R_{C2} of the same defect on the two products are then obtained. We use the hybrid triplet loss to ensure the consistency of R_{C1} and R_{C2} features, while keeping them far apart from the R_A and R_B features.

Furthermore, we obtain the defect-specific features, R_{C1} and R_{C2} , corresponding to the same defect across the two products. To ensure the consistency of these defect features, while simultaneously maintaining their separation from the product features, we employ the hybrid triplet loss strategy. This loss function effectively preserves the similarity between R_{C1} and R_{C2} , reinforcing their alignment, while simultaneously enforcing a substantial distance between them and the product-related features, R_A and R_B . By employing this approach,

we aim to effectively enhance the discriminative power of the extracted features, enabling accurate classification and differentiation of industrial defects on various products within the deep feature space.

E. Overall Pipeline

The complete algorithm of PiCO is depicted in Algorithm 1. In each iteration, the model parameters are updated alternatively through classification and hybrid triplet mining. The iterations involve the processes of *image classification*, *image blending*, and *pixel-level representation*. The product classification and defect classification contribute to the learning of global and local representations, while the hybrid triplet mining aids in acquiring pixel-level representations.

It is worth noting that PiCO is a model-agnostic pretraining method that can be utilized for both CNN and ViT models. By incorporating PiCO into the model pretraining process, better feature representations can be obtained. Rather than proposing a complex neural network architecture, our approach focuses on enhancing the transferability of features through global and local feature learning in the pretraining pipeline. This approach allows for significant performance improvements in downstream industrial tasks without introducing additional model parameters, thereby preserving computational efficiency.

IV. EXPERIMENTS

In this section, we will first introduce the datasets and implementation details. Then, we will conduct ablation studies to validate the effectiveness of each main component of PiCO. Finally, we will report the quantitative comparisons on several downstream industrial benchmarks.

A. Experimental Settings

1) *Dataset and Evaluation Metric*: We study the effectiveness of PiCO on the multilabel defect classification dataset, SewerML [15], and the downstream anomaly detection and localization dataset, MVTec-AD [1]. For evaluation metric, we use the $F1_{\text{normal}}$ score and $F2_{\text{CIW}}$ defect score following SewerML [15]. While on MVTec-AD dataset, we use AUC score as evaluation metric for both of anomaly detection and anomaly localization.

SewerML benchmark evaluates the model performance using two metrics incorporating domain knowledge, based on the $F\beta$ metric [64]

$$F\beta = (1 + \beta^2) \times \frac{\text{Prc} \times \text{Rcll}}{\beta^2 \text{Prc} + \text{Rcll}} \quad (9)$$

where Prc and Rcll are the precision and recall of the classifier, respectively, and β is weighting of recall, such that the recall β times more important than precision.

False negatives in sewer inspections have greater economic consequences than false positives. This is because false negatives can result in undetected faulty pipes, whereas human verification of predicted classes precedes any renovation decisions. Hence, to incorporate this domain knowledge into the evaluation, SewerML benchmark sets $\beta = 2$ when

evaluating the annotated classes. Then, SewerML benchmark introduces $F2_{\text{CIW}}$ metrics to measure the performance using class-importance weights (CIWs)

$$F2_{\text{CIW}} = \frac{\sum_{c=1}^C F2_c \times \text{CIW}_c}{\sum_{c=1}^C \text{CIW}_c} \quad (10)$$

where CIW_c and $F2_c$ are the CIW and $F2$ -score for class c , respectively.

To assess the ability of the tested methods to handle the absence of classes and avoid solely maximizing the $F2_{\text{CIW}}$ score by continuously predicting classes, SewerML benchmark also utilizes the $F1$ -score for normal pipes ($F1_{\text{normal}}$), where we set $\beta = 1$ to evaluate the performance of normal class. The $F2_{\text{CIW}}$ computation excludes normal pipes as the normal case lacks a CIW.

AUC stands for ‘‘Area under the ROC curve.’’ That is, AUC measures the entire 2-D area underneath the entire ROC curve.

2) *SewerML*: dataset focuses on the multilabel defect classification problem and contains 1.3 million images collected over a nine year period. The data are split into a preset training, validation, and test split, containing 1 million, 130k, and 130k images each [25]. The defect classification problem consists of 17 different classes as well as the implicit normal class.

3) *MVTec-AD*: dataset is a comprehensive, multiobject, multidefect industrial anomaly detection dataset with 15 classes. For each anomalous sample in the test set, the ground truth includes both image label and anomaly segmentation.

4) *Implementation Details*: We implement PiCO on Ind-2M dataset with 200 epoch.¹ Following the ImageNet pretraining, we randomly resize images with a ratio of [0.75, 1.25], and crop the images into size of 224×224 . Random horizontal flipping is used for data augmentation. Also, RandAugment is used for color jittering and geometry augmentation. For SewerML fine-tuning, we follow the data augmentation introduced in SewerML paper, i.e., resizing to 224×224 , random horizontal flipping, as well as random color jittering. For the downstream task anomaly detection on MvTec AD dataset, we follow the configuration of uniAD [30], fixing the backbone model, and only training the transformer decoder. The images are resized to 224×224 without data augmentation, and the anomaly location can be obtained by the difference between output features and the fixing backbone features. To ensure a fair comparison between the pretraining methods of ImageNet and SewerML, we have ensured that all hyperparameters in the experiments remain consistent.

B. Ablation Studies

1) *Pretrained Dataset*: In the present study, we aim to explore the influence of utilizing different pretrained datasets for the purpose of industrial image representations. The obtained experimental findings, as outlined in Table II, effectively demonstrate that the adoption of the Ind-2M dataset

¹The epoch here is different from the common definition. We define one epoch as the maximum between the length of product images and the one of defect images.

TABLE II

EXPERIMENTS ARE CONDUCTED WITH ResNET-50 TO VERIFY THE IMPACT OF PRETRAINED DATASET. WE PRETRAIN MODEL ON VARIOUS DATASET, AND FINE-TUNE MODEL ON SEWERML VALIDATION SET

Pre-trained Dataset	$F1_{normal}$	$F2_{CIW}$
ImageNet	81.04	51.55
SewerML (training for scratch)	79.74	50.84
Ind-2M	82.41	51.86

TABLE III

EXPERIMENTS ARE CONDUCTED WITH ResNET-50 TO VERIFY THE IMPACT OF CLASSIFICATION CATEGORIES. WE PRETRAIN MODEL ON VARIOUS CATEGORIES ON IND-2M, AND FINE-TUNE MODEL ON SEWERML VALIDATION SET

Categories	$F1_{normal}$	$F2_{CIW}$
Products	79.91	51.08
Defects	80.95	51.13
Products & Defects	82.41	51.86

for pretraining purposes leads to significantly enhanced performance when compared to alternative pretraining datasets, such as ImageNet and Sewer-ML. Notably, the model that is pretrained on the Ind-2M dataset exhibits noteworthy improvements of 1.37% and 0.31% in relation to the $F1_{normal}$ and $F2_{CIW}$ metrics, respectively, when compared to the model that is pretrained on ImageNet. Furthermore, in terms of the $F1_{normal}$ and $F2_{CIW}$ metrics, the model that is pretrained on Ind-2M outperforms the model that is pretrained on Sewer-ML by 2.67% and 1.02%, respectively.

2) *Classification Categories*: In our study, we investigate the influence of classification categories on industrial image representations, as presented in Table III. The experimental results reveal that the defect categories have a more significant impact compared to the product categories. Specifically, when both classification categories are utilized for pretraining, the model demonstrates superior performance. More specifically, the model pretrained with both product and defect categories exhibits a significant improvement over the model pretrained with product categories alone. This improvement is reflected in the performance metrics $F1_{normal}$ and $F2_{CIW}$, where the former increases by 2.5% and the latter by 0.78%. This indicates that incorporating both product and defect categories during pretraining enables the model to better capture the underlying patterns and features associated with the industrial images. Furthermore, the model pretrained with both product and defect categories also outperforms the model pretrained with defect categories alone. The improvement is evident in the performance metrics $F1_{normal}$ and $F2_{CIW}$, which increase by 1.46% and 0.73% respectively. This suggests that the inclusion of product categories in addition to defect categories enhances the model's ability to discriminate between different types of defects and accurately classify industrial images. Our findings highlight the importance of considering both product and defect categories in the pretraining phase of industrial image analysis models.

3) *Pixel-Level Representations*: In this study, we explore the impact of pixel-level representations on industrial image representations, as depicted in Table IV. The experimental

TABLE IV

EXPERIMENTS ARE CONDUCTED WITH ResNET-50 TO VERIFY THE IMPACT OF PIXEL-LEVEL REPRESENTATIONS. WE PRETRAIN MODEL ON VARIOUS PIXEL-LEVEL SETTINGS ON IND-2M, AND FINE-TUNE MODEL ON SEWERML VALIDATION SET

Pixel-level Representations	$F1_{normal}$	F2
N	82.41	51.86
Y	83.18	53.36

TABLE V

EXPERIMENTS ARE CONDUCTED WITH ResNET-50 TO VERIFY THE IMPACT OF THE WAY OF IMAGE BLENDING. WE PRETRAIN MODEL ON VARIOUS WAYS OF IMAGE BLENDING ON IND-2M, AND FINE-TUNE MODEL ON SEWERML VALIDATION SET

Image Blending	$F1_{normal}$	$F2_{CIW}$
Copy-Paste	82.34	51.93
Poisson Blending	82.99	53.33
CAM-guided Poisson Blending	83.18	53.36

TABLE VI

EXPERIMENTS ARE CONDUCTED WITH ResNET-50 TO VERIFY THE IMPACT OF DILATED CNN. WE PRETRAIN MODEL ON VARIOUS PIXEL-LEVEL SETTINGS ON IND-2M, AND FINE-TUNE MODEL ON SEWERML VALIDATION SET

Dilated CNN	$F1_{normal}$	$F2_{CIW}$
N	82.33	51.74
Y	83.18	53.36

TABLE VII

EXPERIMENTS ARE CONDUCTED WITH ResNET-50 TO VERIFY THE IMPACT OF HYBRID TRIPLET LOSS. WE PRETRAIN MODEL ON VARIOUS PIXEL-WISE LOSS SETTINGS ON IND-2M, AND FINE-TUNE MODEL ON SEWERML VALIDATION SET

Pixel-wise Loss	$F1_{normal}$	$F2_{CIW}$
Softmax CE Loss	80.71	51.07
Hybrid Triplet Loss	83.18	53.36

findings reveal that the model exhibits remarkable performance in multilabel classification when trained using pixel-level supervision. Notably, the model trained with pixel-level supervision surpasses the performance of the model trained without such supervision by 0.77% and 1.5% in terms of $F1_{normal}$ and $F2_{CIW}$, respectively. The results highlight the significance of incorporating pixel-level information in the training process, as it enables the model to capture fine-grained details and spatial relationships within the images. By leveraging pixel-level supervision, the model becomes more adept at distinguishing subtle differences and accurately identifying various defects or anomalies present in industrial images. These findings underscore the importance of considering pixel-level representations when developing models for industrial image analysis tasks.

4) *Image Blending*: In order to investigate the impact of different image blending techniques on industrial image representations, we conducted a comparative analysis as presented in Table V. The evaluated blending methods include copy-paste, Poisson blending, and CAM-guided

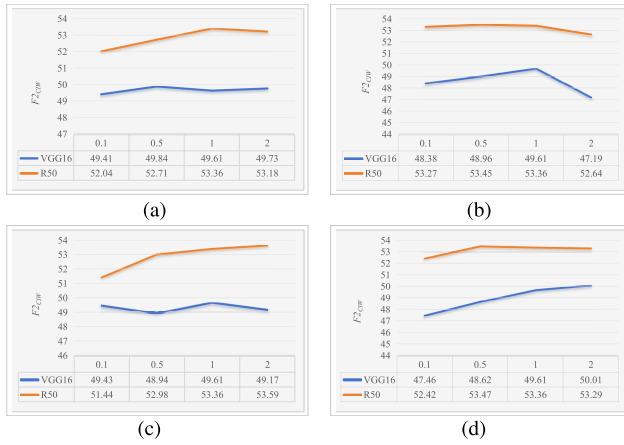


Fig. 5. Curves of the variation of hyperparameter $\lambda_1, \lambda_2, \lambda_3$, and λ_4 under various settings. (a) Curves of the variation of hyperparameter λ_1 . We compare the performance curves of ResNet-50 and VGG-16 under different λ_1 hyperparameter settings, while keeping $\lambda_2 = \lambda_3 = \lambda_4 = 1$. (b) Curves of the variation of hyperparameter λ_2 . We compare the performance curves of ResNet-50 and VGG-16 under different λ_1 hyperparameter settings, while keeping $\lambda_1 = \lambda_3 = \lambda_4 = 1$. (c) Curves of the variation of hyperparameter λ_3 . We compare the performance curves of ResNet-50 and VGG-16 under different λ_1 hyperparameter settings, while keeping $\lambda_1 = \lambda_2 = \lambda_4 = 1$. (d) Curves of the variation of hyperparameter λ_4 . We compare the performance curves of ResNet-50 and VGG-16 under different λ_1 hyperparameter settings, while keeping $\lambda_1 = \lambda_2 = \lambda_3 = 1$.

Poisson blending. Our experimental results demonstrate that CAM-guided Poisson blending exhibits superior performance compared to both direct copy-paste and Poisson blending techniques. Specifically, the model pretrained with CAM-guided Poisson blending achieves notable improvements over the model pretrained with copy-paste in terms of $F1_{normal}$ and $F2_{CIW}$, with gains of 0.84% and 1.43%, respectively. Furthermore, the model pretrained with CAM-guided Poisson blending also outperforms the model pretrained with Poisson blending, yielding enhancements of 0.19% and 0.03% in $F1_{normal}$ and $F2_{CIW}$, respectively. These findings emphasize the significance of the image blending method in enhancing the performance of industrial image representations. The superiority of CAM-guided Poisson blending in comparison to alternative techniques underscores its potential for improving the quality and effectiveness of pretrained models in industrial image analysis tasks.

5) *Dilated CNN*: We also explore the impact of dilated CNNs, as presented in Table VI. By employing dilated convolutions and eliminating the last striding operation in the CNN architecture, the resulting features are able to preserve more fine-grained information. Our experiments demonstrate that the utilization of dilated CNNs can significantly enhance the performance of pixel-level representations. Specifically, we observe improvements of 0.85% and 1.62% in $F1_{normal}$ and $F2_{CIW}$, respectively, when employing dilated convolution in the pretrained model. These findings underscore the effectiveness of dilated CNNs in capturing and leveraging detailed information for industrial image analysis.

6) *Pixel-Wise Loss*: In order to evaluate the impact of the hybrid triplet loss on industrial image representations, we conducted experiments as presented in Table VII. While it may seem intuitive to directly employ the Softmax cross

TABLE VIII
EXPERIMENTS ARE CONDUCTED TO VERIFY THE EFFECTIVENESS OF PiCO ON VARIOUS BACKBONES. WE PRETRAIN MODEL ON VARIOUS BACKBONES WITH DIFFERENT PRETRAINED METHODS, AND FINE-TUNE MODEL ON SEWERML VALIDATION SET

Backbone	Pre-Trained	$F1_{normal}$	$F2_{CIW}$
ResNet-18 [37]	ImageNet	79.32	49.62
ResNet-18 [37]	PiCO	81.85	50.59
ResNet-50 [37]	ImageNet	81.04	51.55
ResNet-50 [37]	PiCO	83.18	53.36
ResNet-101 [37]	ImageNet	82.86	55.37
ResNet-101 [37]	PiCO	84.46	57.45
VGG-16 [65]	ImageNet	78.26	48.14
VGG-16 [65]	PiCO	80.32	49.61
MobileNetV3 [66]	ImageNet	79.69	50.38
MobileNetV3 [66]	PiCO	82.17	52.14
ViT-B [24]	ImageNet	85.35	57.28
ViT-B [24]	PiCO	85.93	58.66

TABLE IX
EXPERIMENTS ARE CONDUCTED WITH ViT-B TO COMPARE THE PRETRAINED METHODS. WE PRETRAIN MODEL WITH VARIOUS PRETRAINED METHODS, AND FINE-TUNE MODEL ON SEWERML VALIDATION SET

Pre-trained Methods	epoch	$F1_{normal}$	$F2_{CIW}$	training time (hours)
ImageNet	200	85.35	57.16	183.4
MAE	800	85.46	56.87	351.1
MoCo v3	200	85.78	57.28	268.8
PiCO	200	85.93	58.66	273.9

entropy loss for pixel-level supervision, this approach faces challenges in effectively distinguishing between intrinsic defects in the product images and blended defects. To address this issue, we opted to utilize the triplet loss as an indirect means of supervising the pixel-level representations. The experimental results demonstrate a significant drop in performance when employing the Softmax cross entropy loss for supervision. This can be attributed to the inherent limitation of classifying intrinsic defects as belonging to the normal class. On the other hand, models pretrained with the hybrid triplet loss exhibit superior performance compared to those pretrained with the Softmax cross entropy loss, achieving improvements of 2.47% and 2.29% in terms of $F1_{normal}$ and $F2_{CIW}$ scores, respectively. These findings highlight the effectiveness of the hybrid triplet loss in enhancing the representation learning process for industrial image analysis.

7) *Hyperparameter λ* : As shown in Fig. 5, we compared the various of the hyperparameter λ in loss weight under different settings. First, as shown in Fig. 5(a), we fixed $\lambda_2 = \lambda_3 = \lambda_4 = 1$ and conducted experimental comparisons with different λ_1 values for VGG-16 and ResNet-50. Through the comparative experiments, we found that for VGG-16, the best performance accuracy can be obtained when $\lambda_1 = 0.5$, while for ResNet-50, $\lambda_1 = 1.0$ yields the best results. Similarly, VGG-16 and ResNet-50 also have different optimal hyperparameters for λ_2, λ_3 , and λ_4 . Therefore, different networks may have different optimal hyperparameters. For simplicity, we maintained the setting of $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 1$ throughout all other experiments.

C. Quantitative Evaluation

1) *Backbones Comparison*: In order to evaluate the performance of the PiCO and ImageNet pretrained models, we conducted experiments using different backbones, namely ResNet-18, ResNet-50, ResNet-101, VGG-16, MobileNetV3, and ViT-B. The results of these experiments are summarized in Table VIII. The findings indicate that the PiCO pretrained model consistently outperforms the ImageNet pretrained model across all backbones and evaluation metrics. This suggests that PiCO has the ability to learn superior industrial representations on the Sewer-ML dataset, regardless of the specific backbone architecture employed.

2) *Comparison of Pretrained Methods*: As presented in Table IX, we conducted a comprehensive evaluation to compare the performance of PiCO with other pretrained methods, namely ImageNet, MAE, and MoCo v3. The evaluation was performed on the Sewer-ML dataset, aiming to assess the effectiveness of PiCO in learning industrial image representations. The experimental results clearly demonstrate that PiCO surpasses the other pretrained methods in terms of performance. This suggests that PiCO has the ability to acquire superior representations for industrial images, surpassing the capabilities of other pretrained methods trained on natural image datasets. Specifically, when compared to ImageNet pretraining, PiCO achieves an improvement of 0.58% and 1.5% in terms of $F1_{\text{normal}}$ and $F2_{\text{CIW}}$ scores, respectively. Moreover, in comparison to MAE pretraining, PiCO exhibits enhancements of 0.47% and 1.79% in $F1_{\text{normal}}$ and $F2_{\text{CIW}}$ scores, respectively. Lastly, when compared to MoCo v3 pretraining, PiCO demonstrates improvements of 0.15% and 1.38% in $F1_{\text{normal}}$ and $F2_{\text{CIW}}$ scores, respectively. These results provide strong evidence that PiCO outperforms other pretrained methods across various evaluation metrics, highlighting its effectiveness in learning representations for industrial images.

In addition, we also compared the training efficiency of our pretraining method with other pretraining methods, as presented in Table IX. It can be observed that the training duration of PiCO is slightly increased compared to ImageNet pretraining, while the training efficiency is similar to that of the MoCo v3 pretraining method. However, the MAE pretraining method requires a larger number of epochs to achieve better pretraining results, resulting in significantly longer training time compared to the PiCO pretraining method.

3) *MvTec AD Detection*: In Table X, we present a comparative analysis of PiCO's performance in the downstream task of anomaly detection. The experimental results demonstrate that PiCO exhibits the capability to enhance the performance of the uniAD method. Notably, the feature representations employed in uniAD are pretrained using conventional recognition tasks, thereby prompting our belief that PiCO has the potential to acquire superior industrial representations in comparison to the ImageNet pretrained models. Remarkably, the uniAD model pretrained with PiCO surpasses its ImageNet-pretrained counterpart by 1.62% in terms of the AUC metric, thereby underscoring the substantial improvement achieved through

TABLE X

EXPERIMENTS ARE CONDUCTED WITH RESNET-101 TO VERIFY THE EFFECTIVENESS OF PiCO ON DOWNSTREAM INDUSTRIAL TASK, ANOMALY DETECTION. WE FIX THE MODEL REPRESENTATIONS AND USE UNIAD FOR ANOMALY DETECTION ON MVTEC-AD DATASET

Methods	Pre-trained	AUC
uniAD	ImageNet	92.2
uniAD	PiCO	93.82

TABLE XI

EXPERIMENTS ARE CONDUCTED WITH RESNET-101 TO VERIFY THE EFFECTIVENESS OF PiCO ON DOWNSTREAM INDUSTRIAL TASK, ANOMALY LOCALIZATION. WE FIX THE MODEL REPRESENTATIONS AND USE UNIAD FOR ANOMALY LOCALIZATION ON MVTEC-AD DATASET

Methods	Pre-trained	AUC
uniAD	ImageNet	95.9
uniAD	PiCO	97.47

the utilization of PiCO in learning representations tailored specifically for industrial anomaly detection tasks.

4) *MvTec AD Localization*: In order to assess the effectiveness of PiCO in the downstream task of anomaly localization, we conducted a comparative analysis, the results of which are presented in Table XI. The experimental findings clearly demonstrate that the utilization of PiCO leads to a notable enhancement in the performance of the uniAD framework. It is worth noting that the feature representations employed in uniAD remain fixed throughout the anomaly detection process. However, we posit that PiCO has the capacity to acquire more refined representations specifically tailored for industrial scenarios, surpassing the capabilities of ImageNet pretrained models. Remarkably, when uniAD is pretrained with PiCO, it achieves a superior performance compared to its counterpart pretrained with ImageNet, exhibiting an improvement of 1.57% in terms of the AUC metric. This substantial advancement further substantiates the efficacy of PiCO in learning representations that are particularly well-suited for the task of industrial anomaly detection.

V. CONCLUSION

In this article, we present the collection of a large-scale industrial production dataset, namely Ind-2M, aimed at facilitating the pretraining of more effective industrial image representations. Ind-2M is specifically curated from industrial scenarios, thereby emphasizing the relevance of the dataset to pretraining models in the industrial domain. Additionally, we introduce a novel pixel-level contrastive pretraining method called PiCO, which is designed to learn industrial image representations. PiCO enhances the global industrial representation through industrial production classification, while simultaneously improving the local industrial representation through a pixel-level self-supervised approach. Experimental results demonstrate the effectiveness of both Ind-2M and PiCO in enhancing industrial image representations. When pretrained on the Ind-2M dataset, PiCO

exhibits strong transferability to tasks such as multilabel defect classification, anomaly detection, and anomaly localization.

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